

Packet-Sequence Modeling for Network Intrusion Detection Using Transformer-based Representation Learning

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March 26, 2026

1 Introduction

Modern network infrastructures have become increasingly complex and indispensable in everyday life, driven by the rapid development of cloud computing, mobile communication, and the proliferation of Internet of Things (IoT) devices. From smart homes and autonomous vehicles to industrial control systems and critical infrastructures, network connectivity is now a fundamental component of modern society. However, this growing dependence on interconnected systems also significantly expands the attack surface of networks.

Cyber attacks such as Distributed Denial-of-Service (DDoS), data exfiltration, ransomware, and advanced persistent threats (APTs) can lead to severe consequences, including service disruption, financial loss, privacy leakage, and even critical infrastructure failure. These risks highlight the urgent need for effective mechanisms to monitor and protect network environments.

Network Intrusion Detection Systems (NIDS) are designed to analyze network traffic and detect malicious or abnormal activities. A NIDS typically inspects packets or flows traversing a network and identifies potential threats based on predefined rules or learned patterns. As a core component of network security, NIDS plays a crucial role in safeguarding modern network infrastructures.

Early NIDS approaches rely on **signature-based detection**, where traffic is matched against known attack patterns [6]. While efficient for detecting known threats, these systems fail to identify unknown or zero-day attacks and require continuous manual updates. To improve adaptability, **machine learning-based approaches** were introduced, utilizing statistical features extracted from network traffic and classifiers such as decision trees, random forests, and support vector machines [7]. Although these methods provide better generalization, they depend heavily on handcrafted features and fail to capture temporal dependencies within traffic.

With the advancement of deep learning, **deep neural network models** such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs) have been applied to intrusion detection [10]. These models enable automatic feature extraction and can capture sequential patterns in traffic data. However, RNN-based approaches suffer from limited parallelism and difficulty modeling long-range dependencies, which restrict their scalability in high-throughput network environments.

More recently, **Transformer architectures** have emerged as a powerful paradigm for sequence modeling due to their ability to capture long-range dependencies through self-attention mechanisms [8]. Transformer-based NIDS frameworks, such as FlowTransformer [1], demonstrate improved performance by learning representations from flow-level sequences. In parallel, **large language models (LLMs)** have achieved remarkable success in natural language processing (NLP), including

tasks such as machine translation, text generation, and reasoning. Recent work such as DoLLM [3] explores applying LLMs to network traffic analysis by treating flows as tokens and leveraging pretrained models (e.g., Llama2) for intrusion detection.

Despite these advancements, most existing Transformer- and LLM-based approaches operate at the **flow level**, where each flow is treated as an atomic unit. This design overlooks an important structural property of network traffic: packets within the same flow naturally form a **temporal sequence**. Compared to flow-level modeling or flow-sequence modeling, packet sequences provide a more fine-grained and natural representation of network behavior.

Transformer architectures are particularly suitable for modeling packet sequences because they:

- effectively process sequential data,
- capture long-range dependencies across packets,
- enable parallel training, overcoming the limitations of RNNs,
- learn complex traffic behavior patterns via attention mechanisms.

More importantly, flow-level aggregation often discards critical information, including:

- inter-packet timing patterns,
- packet length variations,
- role differences between initial and subsequent packets,
- direction changes and burst structures within a flow.

These characteristics can provide important signals for detecting sophisticated or stealthy attacks. Therefore, modeling packet sequences directly represents a promising direction for improving intrusion detection systems.

In this work, we propose a **packet-sequence representation learning framework** for network intrusion detection. Instead of treating flows as single units, we model each flow as an ordered sequence of packets and apply Transformer-based or LLM-based architectures to learn contextual representations. The dataset is constructed from PCAP files using Suricata, which provides packet-level labels for supervised learning.

The main research questions addressed in this work are:

- Can packet-level sequence modeling improve detection performance compared to flow-level approaches?
- Can packet embeddings effectively support both packet-level and flow-level classification tasks?
- Do large language models provide advantages over lightweight Transformer models for network traffic modeling?

2 Related Work

2.1 Overview.

Network intrusion detection has evolved from rule-based systems to data-driven learning approaches. Existing methods can be broadly categorized into signature-based detection, classical machine learning, deep learning, and more recent Transformer and large language model (LLM)-based approaches.

2.2 Signature-based and classical machine learning methods.

Signature-based NIDS represent the earliest and most widely deployed category of intrusion detection systems. Roesch (1999) introduces Snort as a lightweight intrusion detection system that detects malicious traffic using predefined rules and packet inspection [6]. Similarly, Suricata [5] is a high-performance NIDS and network security monitoring engine capable of analyzing both live traffic and PCAP files to generate rule-based alerts. While these systems are efficient for detecting known threats, they are limited by their reliance on predefined signatures and cannot generalize to zero-day attacks.

To address these limitations, classical machine learning approaches were introduced. Sommer and Paxson (2010) analyze the application of machine learning in intrusion detection and highlight that statistical models can improve generalization compared to rule-based systems [7]. In this paradigm, models such as support vector machines (SVM), decision trees, and random forests are trained on flow-level statistical features. Although these methods improve adaptability, they depend heavily on handcrafted features and lack the ability to model temporal dependencies within traffic.

2.3 Deep learning methods.

Deep learning-based NIDS aim to automatically learn feature representations from traffic data. Yin et al. (2017) propose an intrusion detection model based on recurrent neural networks (RNNs), demonstrating that sequence-aware models can improve detection performance [10]. Mirsky et al. (2018) present Kitsune, an online anomaly detection system based on an ensemble of autoencoders, which is designed for efficient and lightweight deployment [4].

Compared with classical machine learning methods, deep learning approaches provide stronger nonlinear modeling capability and reduce the reliance on manual feature engineering. However, many of these models, especially RNN-based architectures, suffer from limited parallelism and difficulty capturing long-range dependencies, which restricts their scalability in high-throughput network environments.

2.4 Transformer and LLM-based methods.

Transformer architectures have recently been introduced into NIDS due to their strong sequence modeling capability. Wu et al. (2022) propose a Transformer-based intrusion detection system (RTIDS), which incorporates positional embeddings to capture sequential relationships among traffic features [9]. The model adopts a stacked Transformer encoder-decoder architecture and is evaluated on the CICDDoS2019 dataset, outperforming traditional machine learning models such as SVM and deep learning models including RNN, FNN, and LSTM.

FlowTransformer further advances this direction by proposing a modular Transformer-based framework for flow-level intrusion detection, where components such as input encoding, backbone architecture, and classification head can be systematically studied [2]. The results demonstrate that Transformer-based models are effective in capturing long-term traffic dependencies.

More recently, large language models have been explored for network traffic analysis. DoLLM reorganizes network flows into flow-sequences and projects them into the semantic space of a pre-trained LLM backbone, showing improved performance in DDoS detection tasks [3]. These approaches indicate that large-scale pretrained models can provide strong representation learning capability for complex traffic patterns.

2.5 Summary of limitations.

Despite the progression from signature-based systems to machine learning, deep learning, Transformer-based, and LLM-based approaches, several limitations remain. Signature-based systems such as Snort and Suricata cannot detect unknown attacks. Classical machine learning methods rely heavily on handcrafted features. Deep learning models improve representation learning but often face scalability challenges.

More importantly, most existing Transformer and LLM-based methods, including RTIDS, Flow-Transformer, and DoLLM, operate at the flow level, treating each flow as an atomic unit. This design overlooks fine-grained packet-level dynamics, such as inter-packet timing, packet length variation, and directional burst patterns within a flow. These limitations motivate our work, which focuses on packet-level sequence modeling to capture richer temporal and structural information for network intrusion detection.

3 Problem Formulation

Let $\mathcal{D} = \{F_1, F_2, \dots, F_N\}$ denote a dataset of network flows, where each flow F_i is composed of a sequence of packets:

$$F_i = \{p_{i,1}, p_{i,2}, \dots, p_{i,T_i}\}$$

Each packet $p_{i,t}$ consists of numerical and categorical features:

$$p_{i,t} = (x_{i,t}^{(num)}, x_{i,t}^{(cat)}), \quad x_{i,t}^{(num)} \in \mathbb{R}^{d_n}, \quad x_{i,t}^{(cat)} \in \mathcal{C}^{d_c}$$

where $\mathcal{C}$ denotes the set of categorical feature domains.

3.1 Label Definition

We consider two types of labels:

- **Flow-level label:**

$$y_i \in \{0, 1, \dots, K_f - 1\}$$

indicating whether a flow is benign or belongs to a specific attack class.

- **Packet-level label:**

$$y_{i,t} \in \{0, 1, \dots, K_p - 1\}$$

indicating whether a packet is malicious or benign.

3.2 Model Formulation

The proposed framework consists of three main components:

1. **Packet Tokenizer**

$$z_{i,t} = f_\theta(p_{i,t}) \in \mathbb{R}^d$$

which maps each packet into a dense embedding.

2. Sequence Encoder

$$h_{i,1:T_i} = g_\phi(z_{i,1:T_i})$$

where g_ϕ is a sequence model (e.g., Transformer or LLM) that captures contextual dependencies across packets.

3. Classification Heads

Flow-level:

$$\hat{y}_i = c_f(\text{Pool}(h_{i,1:T_i}))$$

Packet-level:

$$\hat{y}_{i,t} = c_p(h_{i,t})$$

3.3 Objective Function

Packet-level loss. The packet-level loss is defined over all packets in the dataset:

$$\mathcal{L}_{packet} = \frac{1}{\sum_{i=1}^N T_i} \sum_{i=1}^N \sum_{t=1}^{T_i} \ell(\hat{y}_{i,t}, y_{i,t})$$

where $\hat{y}_{i,t}$ and $y_{i,t}$ denote the predicted and ground-truth labels for packet $p_{i,t}$, respectively.

Flow-level loss. The flow-level loss is defined over all flows:

$$\mathcal{L}_{flow} = \frac{1}{N} \sum_{i=1}^N \ell(\hat{y}_i, y_i)$$

where $\hat{y}_i$ and y_i denote the predicted and ground-truth labels for flow F_i .

3.4 Research Objective

The goal of this study is to learn model parameters (θ, ϕ, c_f, c_p) such that:

- packet-level sequence modeling improves detection performance compared to flow-level models;
- both packet-level and flow-level predictions are supported within a unified framework;
- the model achieves a balance between detection accuracy and computational efficiency.

4 Viable Approach

This section describes a feasible pipeline to implement and evaluate the proposed packet-sequence modeling framework. The pipeline is illustrated in Figure 1.

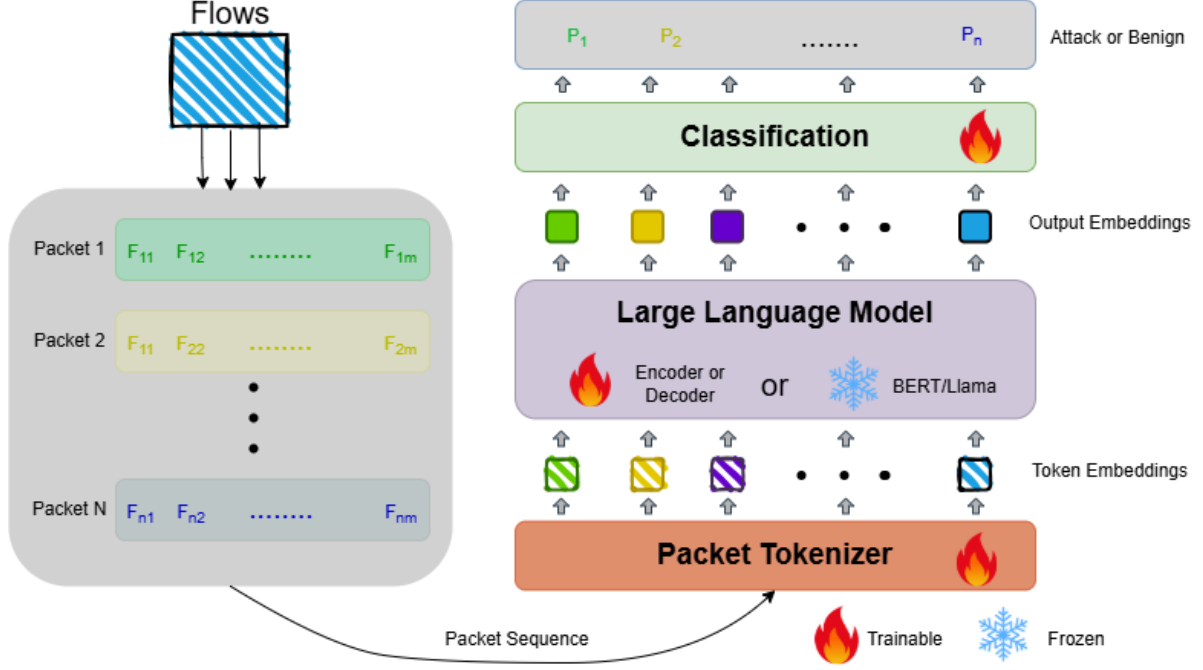


Figure 1: The overall architecture: It consists of three parts: Packet Tokenizer, Sequence Encoder, and Classification Layer.

4.1 Data Collection and Preprocessing

Network traffic data will be collected in PCAP format. Suricata will be used to analyze the PCAP files and generate labeled traffic data. It is important to note that Suricata provides rule-based labels, which may introduce bias and should be treated as pseudo-labels rather than perfect ground truth.

Each flow is extracted and transformed into a sequence of packets, preserving the original temporal order.

4.2 Packet Sequentializer

Packets belonging to the same flow are grouped and ordered by timestamp:

$$F_i = \{p_{i,1}, p_{i,2}, \dots, p_{i,T_i}\}$$

To handle variable-length sequences, padding or truncation will be applied to ensure consistent input sizes.

4.3 Packet Tokenizer

Each packet is converted into a dense embedding using a tokenizer:

- Numerical features are normalized;
- Categorical features are encoded using appropriate strategies depending on their characteristics, such as one-hot encoding or integer encoding;

- Features are concatenated and projected into a fixed-dimensional vector.

4.4 Sequence Encoder

The sequence of packet embeddings is processed by a sequence model:

- **Lightweight Transformer:** used as an efficient baseline;
- **Large Language Model (LLM):** used as a high-capacity model for comparison.

These models capture contextual dependencies across packets within a flow.

4.5 Classification Heads

Two classification tasks are supported:

- Flow-level classification using pooled sequence representations;
- Packet-level classification using token-level outputs.

4.6 Training and Evaluation

The dataset will be split into training, validation, and test sets at the flow level to avoid data leakage.

The following evaluation metrics will be used:

- Accuracy, Precision, Recall, F1-score;
- Area Under ROC Curve (AUROC);
- Inference time for efficiency analysis.

To address class imbalance, techniques such as weighted loss or resampling will be considered.

4.7 Feasibility Considerations

The feasibility of the approach will be evaluated from both performance and efficiency perspectives. In particular, the trade-off between modeling granularity (packet-level) and computational cost will be analyzed to assess its applicability in real-time NIDS scenarios.

5 Initial result of Baselines

As this work is currently at the proposal stage, this section does not report final experimental results. Instead, it presents the baseline models that will be used in the subsequent evaluation and discusses their expected behavior based on their modeling characteristics. The purpose of this analysis is to justify the experimental design and to clarify the comparative setting of the proposed method.

5.1 Planned Baseline Models

To evaluate the effectiveness of the proposed packet-sequence modeling framework, the following baseline models will be considered:

- **Flow-Statistics MLP**: a multilayer perceptron trained on aggregated flow-level statistical features;
- **Random Forest / XGBoost**: traditional machine learning models trained on packet-level or flow-level handcrafted features;
- **LSTM**: a LSTM model for packet-sequence modeling;
- **Small Transformer**: a lightweight Transformer encoder applied to packet sequences;
- **Frozen LLM**: a pretrained LLM backbone combined with an MLP tokenizer and task-specific classification heads.

All baseline models will be evaluated on the same dataset generated from PCAP files using Suricata, which provides packet-level labels. The evaluation will be conducted at both the packet level and the flow level.

5.2 Role of Baselines in This Study

The purpose of these baselines is not only to provide performance references, but also to help analyze the effect of modeling granularity and model complexity. In particular, comparing flow-level statistical models, recurrent sequence models, lightweight Transformers, and LLM-based models will help determine whether packet-level sequence modeling provides measurable advantages for network intrusion detection.

Therefore, the subsequent experimental study will use these baselines to examine whether the proposed method can achieve better packet-level and flow-level detection performance while maintaining a reasonable computational cost.

6 Conclusion

This work proposes a packet-sequence representation learning framework for network intrusion detection. By modeling intra-flow packet sequences, the approach captures fine-grained temporal patterns that are lost in flow-level aggregation. The framework aims to improve detection accuracy while maintaining scalability for deployment in ISP environments.

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